

CARBONMIRROR / ELECTRICAL ARCHITECTURE

DEV-A • 220 VAC nominal / 16 A-class target • FABRICATION HOLD

HAZARDOUS MAINS DOMAIN

INPUT / PROTECTION

Single polarized socket
Fuse + thermal cutoff REQUIRED
Candidates; coordination OPEN

RELAY CONTACTS

K1: 11 → 14 normally open
TE RT314005 candidate
Motor / inrush rating OPEN

METERING: NOT BUILT

ADE9153A hot-domain IC
0.5 mΩ Kelvin-shunt concept
Voltage-divider/filter concept
See pin-level metering design

ISOLATION REQUIRED

ADuM3151 SPI candidate
Separate isolated meter power
No completed daughterboard

PE: continuous protective conductor outside PCB; never switched or bonded to control ground

ISOLATED CONTROL CANDIDATE

ESP32-C3-WROOM-02U

RISC-V edge control + Wi-Fi/BLE
GPIO10 relay / GPIO18 button
GPIO19 status / I²C temperature
External antenna RF review OPEN

LOCAL HARDWARE

Q1 low-side coil driver + D1 flyback
SW1 pushbutton / LED1 indication
TMP102 board temperature only
Independent thermal cutoff REQUIRED

J3 ISOLATED SPI

1:3V3 2:GND 3:SCLK
4:MOSI 5:MISO 6:CS
Never connect HOT GND here

POWER

PS1 IRM-03-5 isolated 5 V candidate
U2 AP63203 → 3.3 V buck
Coil + RF peaks / thermal derating OPEN

END / EDGE / CLOUD

END: measure + actuate within hardware limits
EDGE: local policy, timeout, watchdog, hold
CLOUD: bounded profile and configuration
Offline operation must not depend on cloud
Profiles cannot increase physical current ratings

WHAT IS REAL IN THIS RELEASE

KiCad source, pin-connected schematic, BOM
33 real footprints and package envelopes
Low-voltage routing + actual ERC / DRC
STEP / FCStd / OBJ review geometry
Mains and remaining airwires UNROUTED

RELEASE GATES

Complete mains protection and metrology
Close routing / DRC / independent review
Qualify terminals, copper, PE, creepage
Test EMC, dielectric, thermal and endurance
No Gerber / drill package; never energize

Architecture diagram is a review aid, not a wiring instruction or a safety certification

Sources: Mean Well IRM-03 • TE RT1 • Espressif C3-WROOM-02U • Diodes AP63203 • TI TMP102 • ADI metrology